PROJECT REPORT / CLOUD DETECTION ENGINEERING & THREAT INVESTIGATION

AWS Honeypot Investigation

Cowrie and Dionaea with Wazuh

Author: Sənan Məmmədli | **Report date:** 28 October 2026 | **Scope:** portfolio investigation

Project overview

I built an AWS honeypot environment to collect unsolicited network activity, investigate SSH sessions, and practice the evidence handling used in security operations. Cowrie captured SSH interactions, Dionaea exposed several emulated services, and Wazuh supported searches across imported events. I preserved the sensor archives locally and checked their integrity with SHA-256.

The main finding was repeated SSH activity from **183.23.147.89** with a Go client banner, a HASSH client fingerprint, and an encoded shell probe. Comparing these indicators with IntrusionLabs research produced a meaningful match in tooling and behavior. This is the central investigation in the report.

Findings at a glance

Area	Recorded finding
SSH investigation	Approximately 10.5k events from the leading source in the source-IP chart; 1,307 command-event search results.
Threat intelligence correlation	SSH-2.0-Go, matching HASSH, and the same shell probe described by IntrusionLabs.
Captured file triage	Seven distinct SHA-256 values represented in VirusTotal screenshots: two from Dionaea and five from Cowrie.
Analysis status	Cowrie investigation documented. Dionaea log analysis remains incomplete; its sample assessments are preliminary.

Scope and evidence standard

The report documents observable sensor evidence, external intelligence correlation, and preliminary file triage. Findings are deliberately distinguished from unverified attribution, malware execution, host compromise, or exploit claims.

Reading guide

Deployment and SSH investigation: pp. 2–5. Captured file register and seven file assessments: pp. 6–13. Conclusion and outstanding work: p. 14. References and evidence notes: p. 15.

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Deployment and evidence handling

I separated the two sensors from the Wazuh analysis server. Raw sensor data was exported to Kali Linux, retained in separate archives, and used for later analysis. The workflow below shows the historical import path used for the investigation.

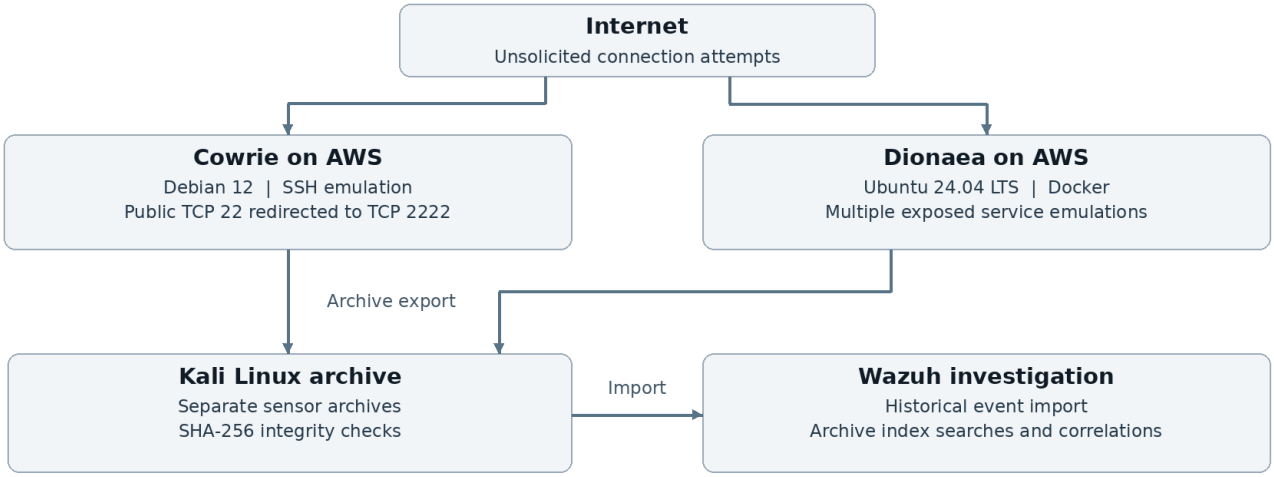


Figure 1. Sensor collection, local preservation, and historical analysis workflow.

Component	Configuration and purpose
Cowrie	Debian 12 on EC2; dedicated **cowrie** user and systemd service. JSON events and session artifacts supported SSH analysis.
Dionaea	Ubuntu 24.04 LTS on EC2; `dinotools/dionaea:nightly` managed with Docker Compose. Bind mounts under /srv/dionaea retained data outside the container.
Wazuh	Separate EC2 analysis server. Cowrie investigation used `wazuh-archives-*` and imported source `/srv/cowrie-import/events.json`.
Kali Linux	Local archive storage and manual investigation. Separate Cowrie and Dionaea .tar.gz archives passed `sha256sum -c` verification.

Network and administrative access

Cowrie listened on TCP 2222, with public TCP 22 redirected to it. The real administrative SSH service moved to TCP 22222 and was restricted to my administrative IP. Dionaea used separate administrative SSH access restricted by source IP. Exposed services included FTP 21, HTTP 80, MSRPC 135, SMB 445, MSSQL 1433, MQTT 1883, MySQL 3306, Memcached 11211, and MongoDB 27017.

Archive checksums support change detection after collection and transfer; they do not prove the authenticity of every event or establish a complete forensic chain of custody.

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SSH activity identified in Wazuh

The source-IP distribution made **183.23.147.89** the first investigation priority. Its bar represents approximately 10.5k events in the displayed query. Since an SSH session creates several event types, this is event volume, not a count of attacks or distinct sessions.

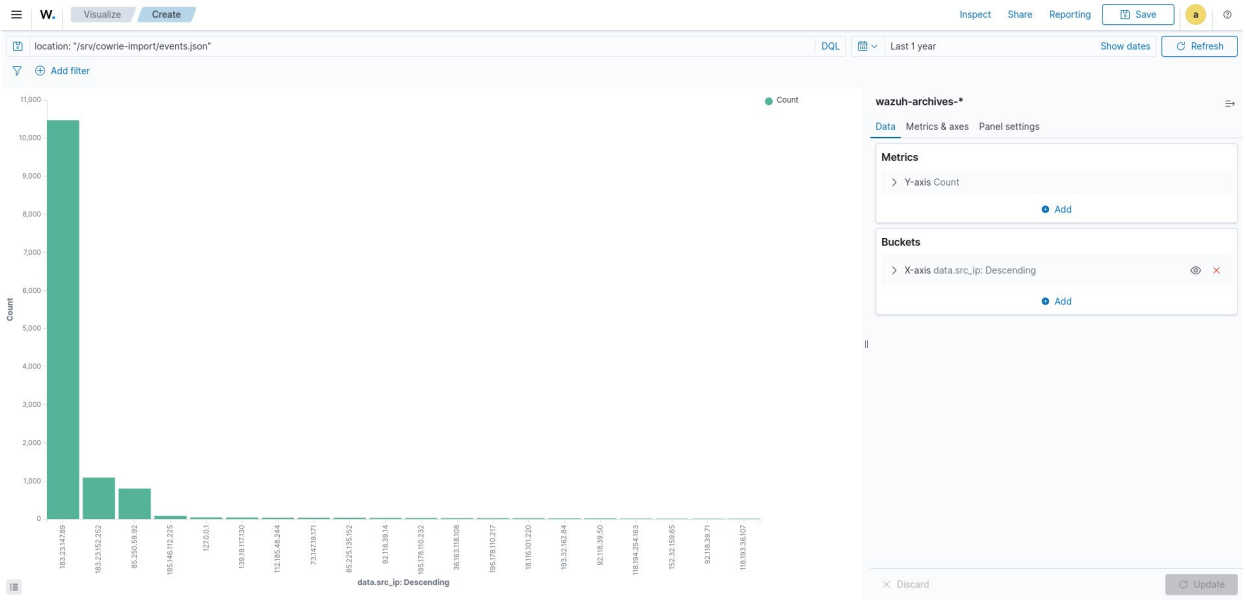


Figure 2. Cowrie events grouped by source IP; the leading bar represents approximately 10.5k events.

Query / observation	Recorded result	Interpretation
Source IP	183.23.147.89	Leading source in displayed chart
Command events	1,307 hits	cowrie.command.input search total
Client version events	1,310 hits	Inspected record contains SSH-2.0-Go
Key exchange events	1,309 hits	Inspected record contains HASSH fingerprint
Original event date	24 September 2026	Visible in selected sensor records

Session behavior and authentication

The event-type chart in the original project analysis showed roughly similar counts for connection, client version, key exchange, accepted login, session parameters, command input, and session closure. This supports a repeated automated interaction pattern. Exact sequencing and duration require grouping by session ID and sorting by original event timestamp.

A `cowrie.login.success` event means the honeypot accepted credentials under its configured rules. It does not establish compromise of the administrative SSH service or underlying EC2 host.

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Shell probe and IntrusionLabs correlation

The command-event search returned **1,307 results**. In the displayed 500-record field summary, each loaded value was the same shell probe:

```
echo -e "\x6F\x6B"
```

These hexadecimal bytes represent “o” and “k”; with escape processing enabled, the command prints “ok”. This is consistent with a shell-response check. The screenshot captures command input, not actual command output.

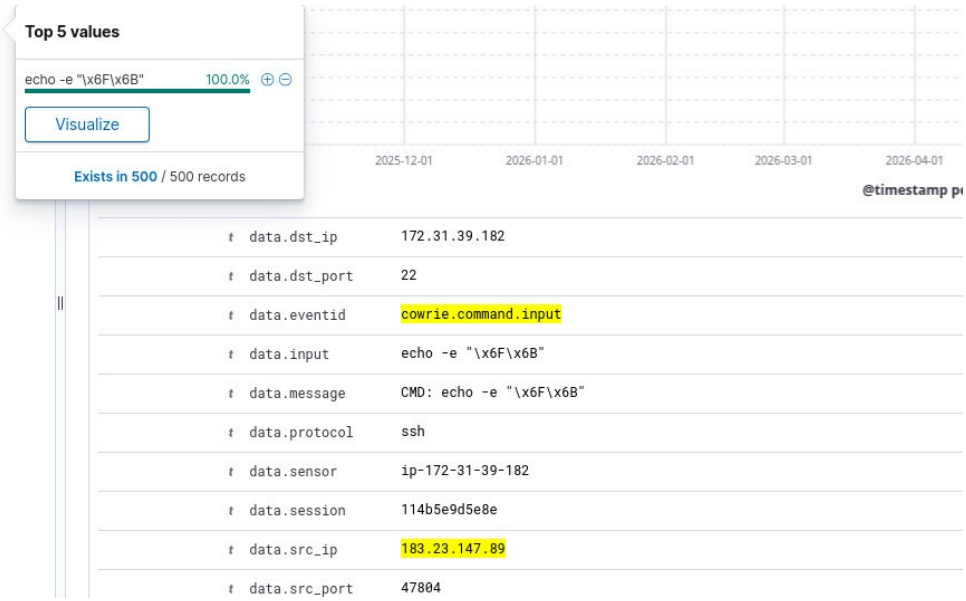


Figure 3. Wazuh command-event field summary and inspected record. The 100% statistic covers 500 loaded records, not necessarily all 1,307 results.

Comparison with published research

IntrusionLabs described a Go-based SSH brute-forcer in a 2 May 2026 investigation, updated on 4 May. Three indicators match the published report. [R1]

Indicator	Local observation	IntrusionLabs comparison
Source IP	183.23.147.89	185.246.152.229; different source
Client banner	SSH-2.0-Go	Same banner
HASSH	01ca35584ad5a1b66cf6a9846b5b2821	Same fingerprint
Shell probe	echo -e "\x6F\x6B"	Same command
Credentials	Not established by shown records	root and distinct passwords reported

Assessment

These matches support suspected automated Go SSH credential-testing behavior, consistent with the tooling class reported by IntrusionLabs. They do not prove a shared operator, identical binary, or common campaign. “Melbikomas” refers to infrastructure discussed in the external article and is not linked to my observed source IP. [R1]

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Fingerprint evidence and investigation method

Two enlarged excerpts from the original Wazuh records show the observed client banner and the key-exchange fingerprint. Cropping removes unrelated interface space without changing the visible values.

CLIENT VERSION EVIDENCE

@timestamp per week

🗓	data.timestamp	Sep 24, 2026 @ 21:59:27.411
t	data.uuid	a586c5a2-b4b4-11f1-a655-06ffd8267ea9
t	data.version	SSH-2.0-Go
t	decoder.name	json

Figure 4. Inspected client-version record: original sensor time, SSH-2.0-Go, and event context.

KEY EXCHANGE EVIDENCE

t	data.eventid	cowrie.client.kex
t	data.hassh	01ca35584ad5a1b66cf6a9846b5b2821
t	data.hasshAlgorithms	curve25519-sha256@libssh.org,ecdh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffssh.com,chacha20-poly1305@openssh.com,arcfour256,arcfour128,arcfour,aes128-cbc,3des-cbc,ae2-256,hmac-sha1,hmac-sha1-96;none

Figure 5. Inspected key-exchange record: HASSH value and part of the negotiation algorithm context.

HASSH summarizes ordered SSH algorithm negotiation lists. It can compare client implementations and configurations, but cannot uniquely identify an operator or executable. The MD5 HASSH fingerprint is distinct from SHA-256 file-integrity checksums. [R2]

Reproducible search starting point

In `wazuh-archives-*`, start with: `location: "/srv/cowrie-import/events.json"` and `data.src_ip: "183.23.147.89"`. Extend with `data.eventid` for `cowrie.command.input`, `cowrie.client.version`, or `cowrie.client.kex`. Complete aggregations and distinct `data.session` counts are required before treating a pattern as uniform.

Field note: The inspected key-exchange screenshot displays the fingerprint under `data.hash`; field mappings should be checked against the actual imported dataset.

Time interpretation: The raw banner record contains `2026-09-24T17:59:27.411252Z`; the interface shows 21:59:27.411 (UTC+4). Imported index timestamps may differ from the source event time.

Proposed detection: Correlate repeated authentication attempts, client version, HASSH, and probe by source and session, then assess rate. This is a detection idea, not an implemented or validated Wazuh rule; banner or fingerprint alone is insufficient for blocking.

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Captured file register and analysis scope

Files were collected through both honeypots and investigated using VirusTotal as an enrichment source. The seven screenshots represent **seven distinct SHA-256 values**: two from Dionaea and five from Cowrie. These records are a selected evidence set, not an exhaustive payload inventory.

ID	Sensor	SHA-256 prefix	Type / size	VT result
D1	Dionaea	9348ff359030	Not visible	Not visible
D2	Dionaea	42a5ae53dcdb	PE DLL / 5.02 MB	68 / 72
C1	Cowrie	c81b72a01d7d	ELF 64-bit / 4.25 MB	38 / 64
C2	Cowrie	94f2e4d8d443	ELF 64-bit / 28.90 MB	47 / 64
C3	Cowrie	6869ce81729a	ELF 64-bit / 896 KB	39 / 64
C4	Cowrie	47b268c21591	ELF 64-bit / 160 KB	34 / 64
C5	Cowrie	a8bf9a1f5ec1	ELF 64-bit / 2.91 MB	39 / 63

VT result = number of vendors flagging the sample / number shown at the time of the screenshot. These are historical observations, not refreshed verdicts.

What the triage supports

The screenshots preserve hash identifiers, visible type and size information, vendor ratios, automated labels, and rule matches. Dionaea samples have WannaCry-associated labels; Cowrie samples have miner or trojan labels, frequently including “multiverze”. These indicate investigation priorities rather than independently verified family or campaign attribution. [R3]

Limitations and outstanding verification

Dionaea log analysis is incomplete: the original connection, transfer path, and exploit sequence have not been reconstructed. For all seven samples, the next evidence step is to verify local hashes and link each artifact to its sensor timestamp, connection or session ID, transfer event, and available download URL.

Some VirusTotal filenames contain “sftp”, “root”, and “sshd”. These may reflect earlier submissions, not local capture time, transfer method, or an installed daemon. Cowrie upload and download event types should be examined directly. [R4]

Important separation: No session-level link has been established between these seven payloads and the SSH-Go case involving 183.23.147.89. Collection alone does not prove execution, persistence, mining, encryption, or compromise of the host.

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Dionaea sample D1

WannaCry-associated classification

Detection ratio / type / size: not visible in screenshot

SHA-256

`9348ff35903052b6874508e0fa116dacc4dca5b0a60710a6084c3ac5f01ae85f`

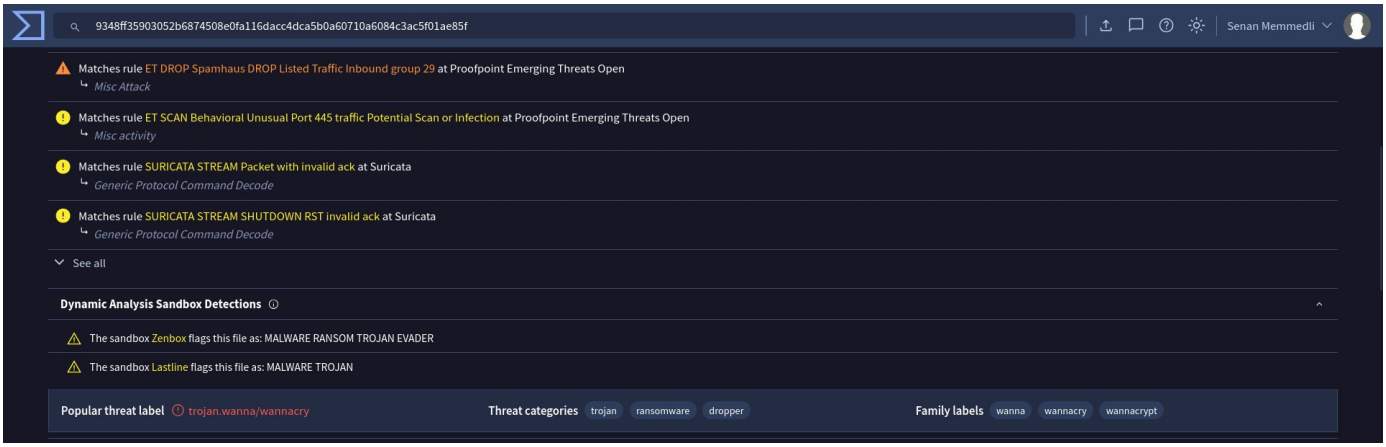


Figure 6. VirusTotal record showing WannaCry-associated classifications and sandbox/network detections.

What the screenshot shows

VirusTotal displays **trojan.wanna/wannacry** with trojan, ransomware, and dropper categories. Visible sandbox verdicts include malware and ransomware-related classifications. Network-rule matches include unusual TCP 445 activity, Spamhaus-listed traffic, and Suricata stream anomalies.

Assessment

The external results warrant prioritizing this file as a suspected WannaCry-associated payload. The TCP 445 match is relevant to an SMB investigation but comes from VirusTotal analysis; it is not evidence that this alert fired in Dionaea, that a specific vulnerability was exploited, or that local files were encrypted.

View VirusTotal file record ↗

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Dionaea sample D2

DLL with WannaCry rule match

68 / 72 vendors · PE DLL · 5.02 MB

SHA-256

`42a5ae53dcdbe4db925440b319fbd3e46be2a78e214b9b41e9b9dd293415ee13`

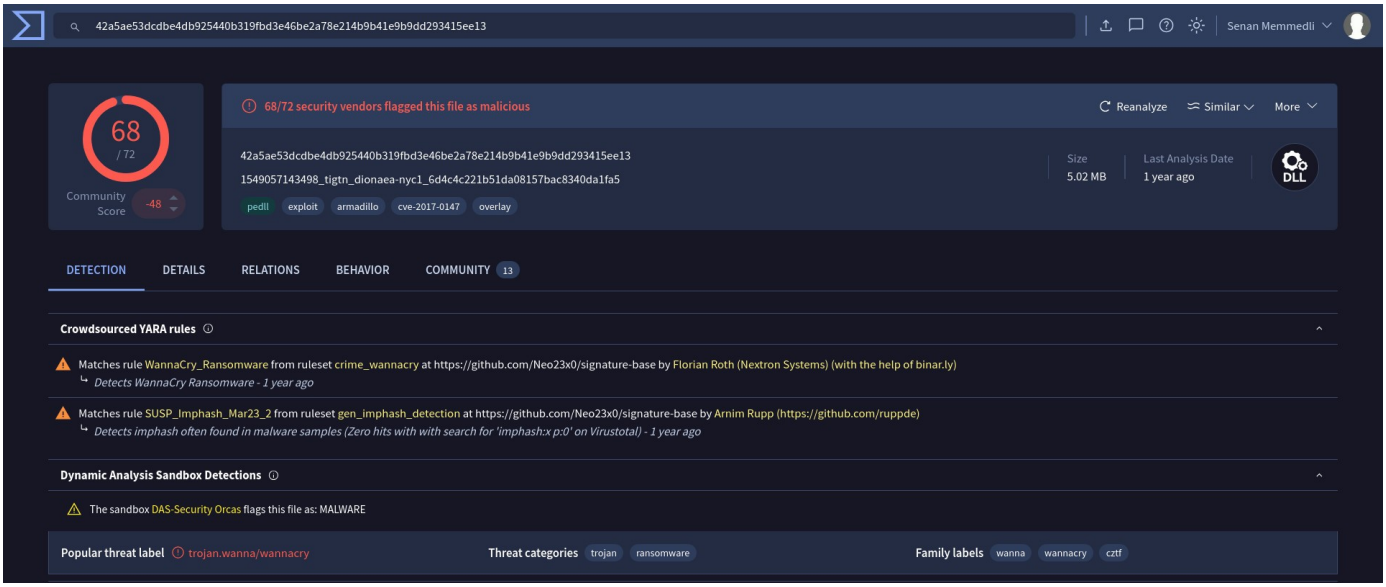


Figure 7. VirusTotal record: antivirus detection, WannaCry rule match, and additional metadata.

What the screenshot shows

The screenshot shows broad antivirus detection, a **WannaCry_Ransomware** YARA match, a suspicious import-hash rule, and a sandbox malware verdict. The popular label is trojan.wanna/wannacry; metadata includes exploit and CVE-2017-0147 tags.

Assessment

The combined vendor results and WannaCry-specific rule justify high-priority triage. The CVE tag is external metadata, not a local exploitation trace. It does not prove CVE-2017-0147 was used against the sensor and should not be converted into an EternalBlue claim without protocol-level validation.

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Cowrie sample C1

ELF with mining indicators

38 / 64 vendors · ELF 64-bit · 4.25 MB

SHA-256

`c81b72a01d7d2a0eaa25101b6d24cf73128b4fd545a59817c0f2373087162538`

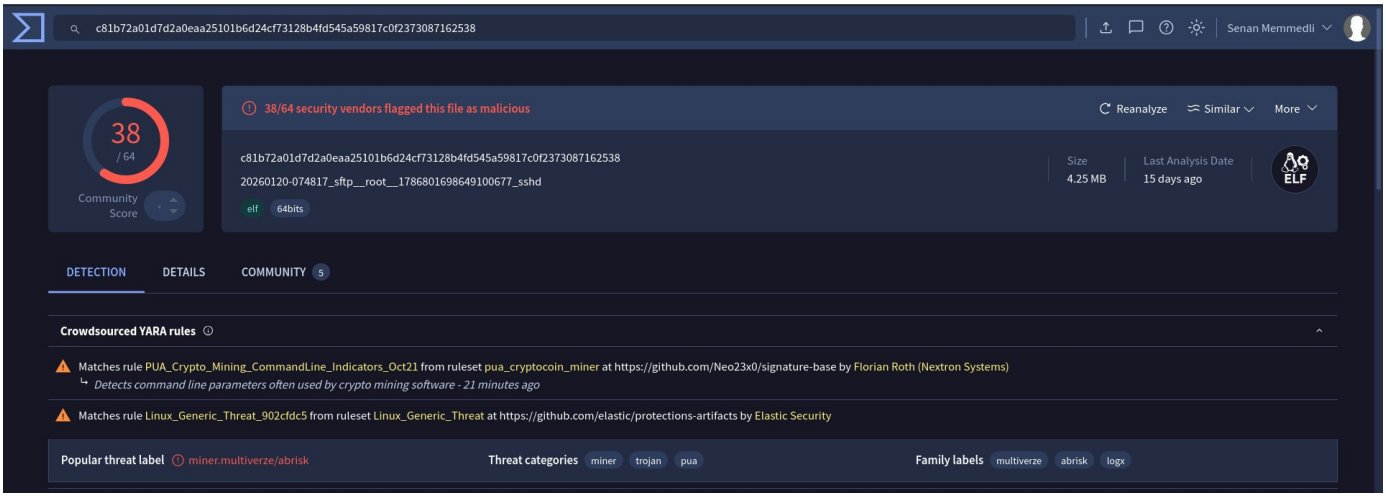


Figure 8. VirusTotal record for a Linux ELF file with miner-related labels and YARA indicators.

What the screenshot shows

VirusTotal shows **miner.multiverze/abrisk** alongside miner, trojan, and potentially unwanted application categories. A YARA rule identifies command-line mining indicators; another identifies a generic Linux threat. The displayed submission name ends in “sshd”.

Assessment

These observations support a mining-related triage hypothesis, not confirmed mining on the sensor, a specific coin or pool, or replacement of the real SSH daemon. “sshd” is a possible masquerading lead; local file location and execution context would be needed to establish it.

View VirusTotal file record ↗

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Cowrie sample C2

ELF with mixed signature matches

47 / 64 vendors · ELF 64-bit · 28.90 MB

SHA-256

`94f2e4d8d4436874785cd14e6e6d403507b8750852f7f2040352069a75da4c00`

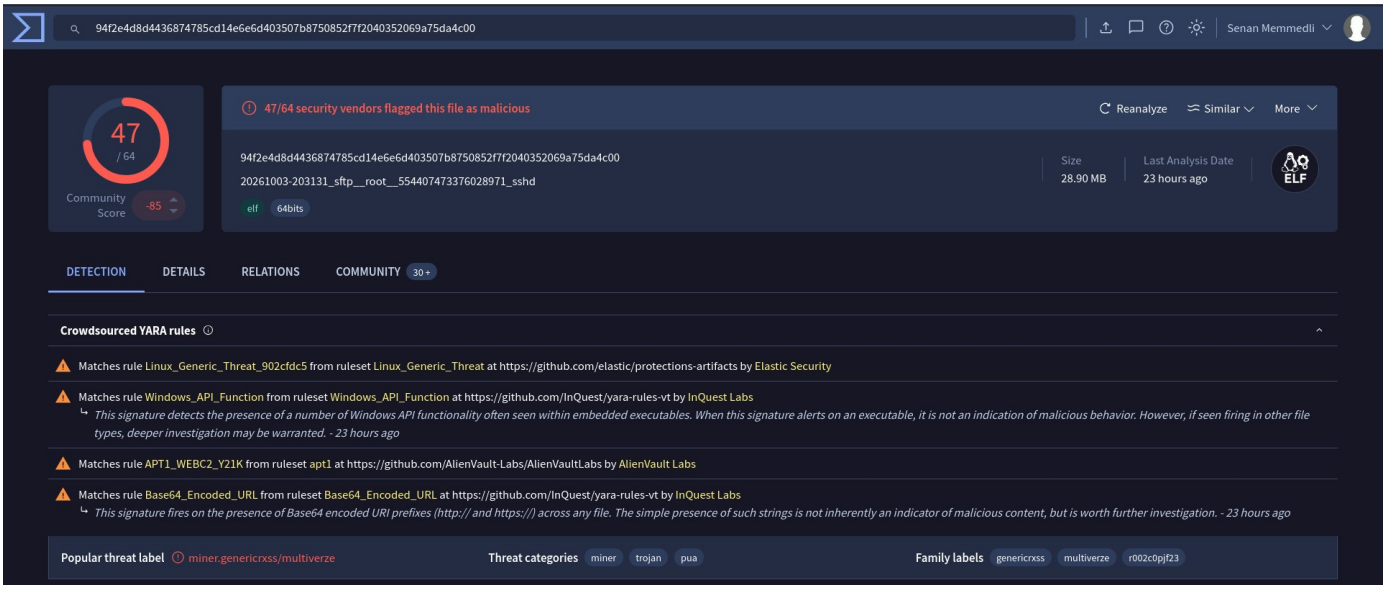


Figure 9. VirusTotal record with generic Linux and mixed platform/signature matches.

What the screenshot shows

Popular label: **miner.genericrxss/multiverze**. Visible YARA matches include a generic Linux threat rule, Windows_API_Function, APT1_WEBC2_Y21K, and Base64_Encoded_URL. The screenshot itself cautions that some patterns are not inherently malicious.

Assessment

Vendor detection supports further investigation, but each rule must be interpreted separately. Windows-related strings do not change the file's ELF format. A rule named after APT1 does not establish APT1 attribution, and a Base64 URL match does not prove an operational command-and-control channel.

View VirusTotal file record ↗

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Cowrie sample C3

ELF with conflicting automated assessments

39 / 64 vendors · ELF 64-bit · 896 KB

SHA-256

`6869ce81729acda83597601eb7c89c0bdda23d41bd5fe6900256dfb389a9bf47`

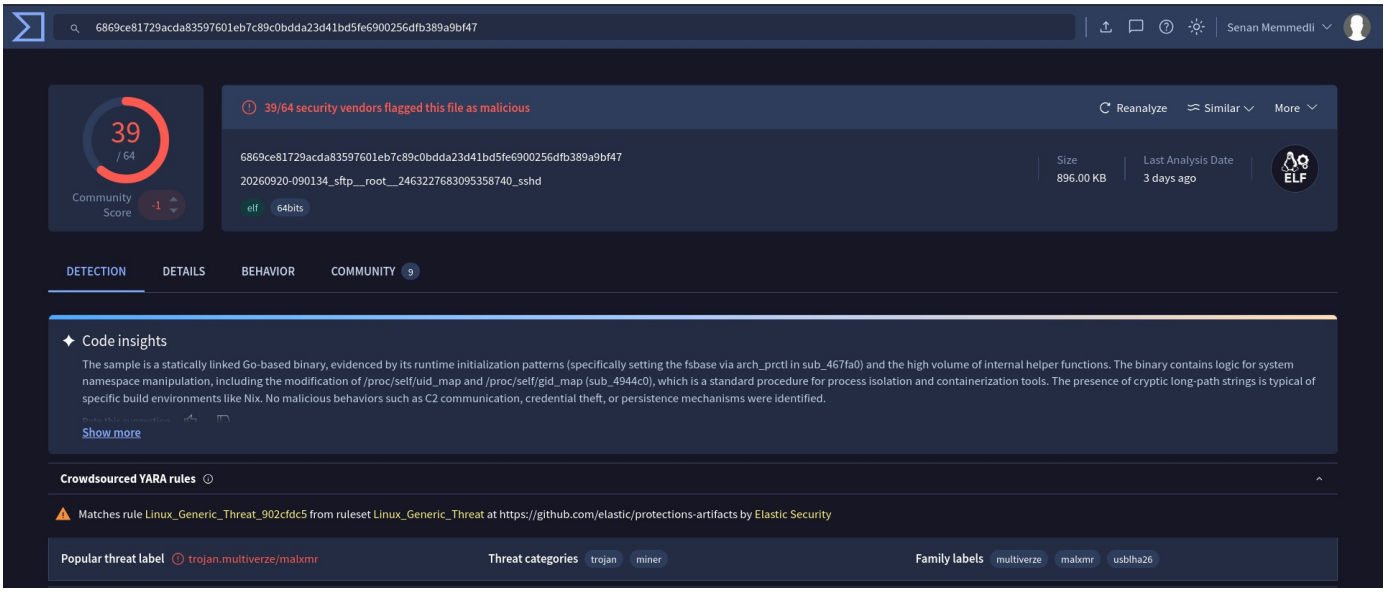


Figure 10. VirusTotal record with conflicting vendor labels and automated Code Insights.

What the screenshot shows

VirusTotal shows **trojan.multiverze/malxmr** and a generic Linux threat rule. Its automated Code Insights describes a statically linked Go binary and user-namespace manipulation, but does not identify C2, credential theft, or persistence in the analyzed code.

Assessment

The disagreement between vendor labels and automated code summary remains unresolved. The summary does not prove the file benign, nor does it validate every antivirus label. Namespace manipulation and Go runtime patterns can appear in legitimate software; provenance and behavior require independent examination. [R5]

View VirusTotal file record ↗

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Cowrie sample C4

ELF with suspected obfuscation

34 / 64 vendors · ELF 64-bit · 160 KB

SHA-256

`47b268c21591069bfe4099833ad66b8138a53ab2dcb866e040d466aee1f8624c`

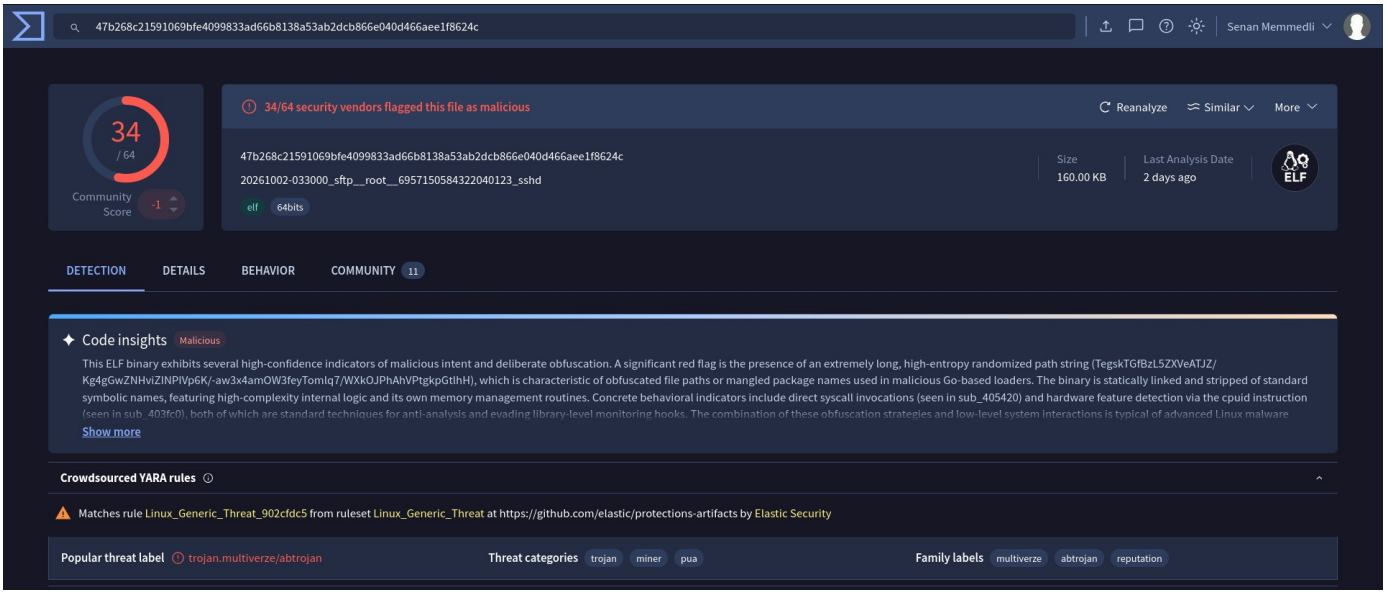


Figure 11. VirusTotal record indicating generic Linux detections and automated anti-analysis hypotheses.

What the screenshot shows

The screenshot shows **trojan.multiverze/abtrojan**, a generic Linux threat rule, and an automated malicious verdict. Code Insights describes a stripped, statically linked binary, unusual path strings, direct system calls, and CPU feature checks, interpreting these as possible obfuscation or anti-analysis signals.

Assessment

The automated interpretation is a hypothesis rather than an established behavior. Stripping, direct system calls, and CPU feature checks are not exclusive to malware. With the vendor flags and honeypot context, they justify examination but not a specific loader-family or evasion claim. No independent reverse-engineering result is claimed.

View VirusTotal file record ↗

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Cowrie sample C5

ELF with suspected remote-access capabilities

39 / 63 vendors · ELF 64-bit · 2.91 MB

SHA-256

`a8bf9a1f5ec1502b8f391c49f533d336e05f39f3a20e5fa1dafd796bceadf48d`

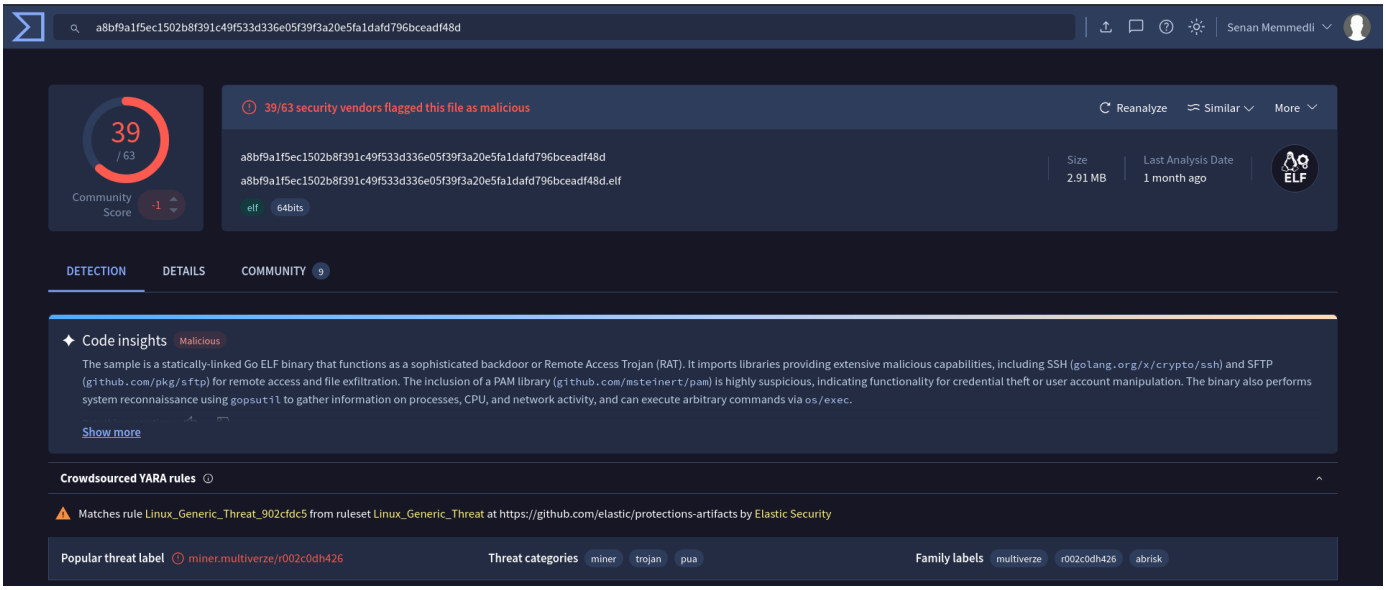


Figure 12. VirusTotal record with mining labels and an automated backdoor-capability narrative.

What the screenshot shows

VirusTotal shows **miner.multiverze/r002c0dh426** and a generic Linux threat rule. Code Insights describes a possible Go backdoor or remote-access trojan, citing SSH/SFTP libraries, a PAM library, system-information functions, and command execution via `os/exec`.

Assessment

The miner label and backdoor narrative suggest different possible capabilities. Neither proves these actions occurred in this project. Library imports do not establish credential theft, exfiltration, or actual use. “r002c0dh426” is retained as a detection label, not independently verified as a malware family. A Go-related description does not connect this file to the SSH-Go investigation.

View VirusTotal file record ↗

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Conclusion and future work

This investigation combined an AWS-based honeypot environment with centralized log search, preserved sensor evidence, external threat intelligence, and preliminary captured-file triage.

Established findings

Workstream	Outcome supported by the evidence
AWS infrastructure	Separate EC2 sensor and analysis roles; Cowrie and containerized Dionaea; restricted administrative access.
Cowrie SSH analysis	Repeated activity from 183.23.147.89, with 1,307 command-event query hits and a consistent probe in the 500-record summary.
Threat intelligence	SSH-2.0-Go, HASSH, and a shell probe match reported Go SSH credential-testing behavior; campaign identity is not established.
Captured file triage	Seven selected SHA-256 values documented and reviewed with VirusTotal, without claiming local execution or host compromise.
Evidence handling	Sensor archives preserved locally and checked with SHA-256; complete chain of custody not claimed.

Outstanding investigation work

- Complete Dionaea connection and service-log analysis, including the original delivery context for the two files.
- Link each captured file hash to the corresponding transfer event, sensor timestamp, and session or connection ID.
- Run full, session-level aggregations for repeated SSH probe behavior and source activity over original event timestamps.
- Develop and evaluate a multi-signal detection approach before describing it as an implemented Wazuh rule.

Background threat descriptions: proper interpretation

Microsoft's general AndroidOS/Multiverze and Linux/Multiverze descriptions [R6–R7] provide terminology and investigation hypotheses, but are not analyses of these captured hashes. They do not establish Android targeting, SSH compromise, a particular bot, framework exploitation, DDoS activity, or persistence in the local environment. The shared "Multiverze" token alone cannot group these files into one family or campaign.

Final assessment

The central case is evidence-based correlation of automated SSH behavior with published research. The selected malware results enrich follow-up investigation but remain preliminary. The project demonstrates a repeatable workflow for cloud deployment, log-driven prioritization, evidence preservation, careful attribution, and documentation of uncertainty.

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Sources and evidence notes

External references below were listed in the source report as checked on 28 October 2026. Original local evidence includes the university report, Wazuh screenshots, and seven VirusTotal file-record screenshots. The full monitoring window and sensor-wide totals are not claimed.

[R1] IntrusionLabs — One IP, 23,307 unique passwords, eight hours: anatomy of a “rise” that wasn’t (2 May 2026, updated 4 May).

[R2] Salesforce HASSH — SSH fingerprinting project.

[R3] VirusTotal — popular_threat_classification documentation.

[R4] Cowrie — Output Event Code Reference.

[R5] VirusTotal Blog — Exploring the VirusTotal Dataset: An Analyst’s Guide to Effective Threat Research.

[R6] Microsoft Security Intelligence — Trojan:AndroidOS/Multiverze.

[R7] Microsoft Security Intelligence — Trojan:Linux/Multiverze.

Screenshot-to-file identity mapping

ID	Original screenshot filename	Source
D1	9348.jpg	Dionaea
D2	42a_dio.jpg	Dionaea
C1	c81.jpg	Cowrie
C2	94f2.jpg	Cowrie
C3	6869c.jpg	Cowrie
C4	47b.jpg	Cowrie
C5	a8bjpg	Cowrie

Evidence interpretation notes

All VirusTotal screenshots reflect the displayed result at the time they were captured; ratios and labels can change. Relative “last analysis” times and dates embedded in submitted filenames are not local capture timestamps. File links in the sample pages resolve to the individual SHA-256 records.

Automated summaries and sandbox observations are attributed to their external sources. No independent malware execution or reverse engineering is claimed. A honeypot login, file capture, signature match, or vendor verdict alone does not establish infection of the real EC2 host.